DEPARTMENT OF INFORMATION TECHNOLOGY

Windows Server Administration

Practical Assessment Assignment-1

VirtualBox | Windows Server 2019 | Windows 10 / 8.1

Student Name		**Date**	
Registration No.		**Time**	
Section / Batch		**Marks**	/ 100

GENERAL INSTRUCTIONS

Duration: 2.5 to 3 Hours | Total Marks: 100 | All tasks are compulsory.

1. Perform all tasks sequentially. Do not skip any task — later tasks depend on earlier configurations.

2. Take a screenshot immediately after completing each task. Screenshots taken out of sequence will not be accepted.

3. For every task that asks you to 'explain' or 'document', write your answer in your own words. Copied text from any source will receive zero marks.

4. A detailed lab report must be submitted at the end. The report structure and requirements are defined in the final section of this document.

5. Both virtual machines must be running simultaneously on VirtualBox throughout the assessment.

6. Any task involving a deliberate misconfiguration must be fully restored before proceeding to the next task.

SECTION 1 — VirtualBox Environment Verification & Baseline Documentation

Marks: 10 | Estimated Time: 20 minutes

1. Before powering on either virtual machine, open VirtualBox and access the settings of both VMs. Navigate through the **System**, **Network**, and **Display** tabs for both machines. Create a structured comparison table in your report documenting every configuration option visible, and identify at least three hardware-level differences between your server VM and your client VM.
2. Both VMs are currently assigned to the same Internal Network adapter in VirtualBox. Without changing any setting inside the VMs, temporarily change one VM's adapter to NAT and the other to Internal Network. Attempt to power on both, run **`ipconfig`** on each, and document the result with screenshots. After testing, restore the correct Internal Network configuration on both machines and explain in writing why this experiment failed.
3. Power on both VMs simultaneously. Open Task Manager on your **host machine** and document RAM, CPU, and disk consumption while both VMs are running. Then shut down one VM and document the change in host resource usage. Calculate the exact difference in each resource category and write a short paragraph explaining what these figures mean for real-world virtualization environments.
4. While both VMs are running, find the MAC address of each VM's network adapter from the VirtualBox main window (not from inside the VM). Then, inside each VM, open Command Prompt and verify the MAC address using **`ipconfig /all`**. Confirm both match, document the results in a table, and explain in your own words what a MAC address is, how it differs from an IP address, and why every device on a network must have a unique MAC address.
5. Take a named snapshot of both machines in their current unconfigured state. Then intentionally misconfigure any one setting on the server — document what you changed and why. Take a second snapshot of the broken state. Restore the machine to the first snapshot, confirm everything is back to normal with a screenshot, and write a paragraph explaining why snapshots are a critical feature in production server environments and when an IT administrator would use them.

> **NOTE:** Changing the network adapter type between tasks may affect connectivity in later sections. Always verify network settings are correctly restored before proceeding.

SECTION 2 — Windows Server 2019 Deep Configuration

Marks: 25 | Estimated Time: 40 minutes

1. Open Server Manager on Windows Server 2019. Before making any change, take a full screenshot of the **Local Server** tab. In your report, create a table listing every single field visible on this tab — document the current value of each field and a one-sentence explanation of what that field represents.
2. Research and document the official Windows Server hostname naming conventions: maximum character length, allowed characters, and prohibited characters. Then rename your server to a professional hostname following these rules. Separately, attempt to enter a hostname that **deliberately violates** at least one rule — screenshot the error message produced, and after documenting it, apply the correct name and restart.
3. After the server restarts, open Command Prompt and run each of the following commands one by one. For each command, paste its complete output into your report and write a clear explanation of what that command reveals about system identity:
    a. **`hostname`**
    b. **`whoami`**
    c. **`systeminfo | findstr /C:"Host Name"`**
    d. **`systeminfo | findstr /C:"OS Name"`**
    e. **`systeminfo | findstr /C:"System Boot Time"`**
4. Before changing the time zone, run **`w32tm /query /status`** in Command Prompt and document the full output. Change the time zone to your correct local region, then run the same command again. Compare both outputs in a table and explain the role of time synchronization in a networked server environment — specifically what goes wrong when server and client clocks are out of sync.
5. After setting the correct time zone, deliberately set the system clock 30 minutes ahead of actual time. Run **`w32tm /query /status`** again and document the output. In your report, write a detailed paragraph explaining why incorrect system time on a server can cause critical failures in authentication, event logging, and scheduled tasks. Restore the correct time before proceeding.
6. Assign the static IP address **`192.168.10.1`** with subnet mask **`255.255.255.0`** to the server's network adapter. After applying, run the following four commands and document the complete output of each — for each command, annotate your output by explaining what every individual field or line means:
    f. **`ipconfig`**
    g. **`ipconfig /all`**
    h. **`netsh interface ip show config`**
    i. **`netsh interface ip show address`**
7. Open the network adapter's Properties window on the server. Without changing anything, document every protocol and service listed in the adapter's component list. Select at least three of them and write a detailed explanation of what each one does and why it matters in a server environment.
8. Attempt to add a **second static IP address** on the same network adapter using the Advanced TCP/IP settings (do not replace the first IP — add it as an additional address). Document whether this is possible, show the result with screenshots, and explain a real-world scenario where a server would require multiple IP addresses on a single adapter. Remove the second IP and restore the single static configuration before proceeding.

9. Run **`ipconfig /all`** on the server and locate the **DNS Suffix** field. Document its current value and write an explanation of what a DNS Suffix is, why it appears in the output, and what role it would play once the server is promoted to a Domain Controller.

> **NOTE:** All static IP configurations applied in this section will be used as the basis for connectivity testing in Section 4. Any misconfiguration here will directly cause failures in later sections.

SECTION 3 — Windows 10/8.1 Client Deep Configuration

Marks: 20 | Estimated Time: 30 minutes

1. On the client machine, run `ipconfig /all` before making any configuration changes. Document the complete output. If an automatic IP address has been assigned, identify whether it falls within the APIPA range — document what APIPA stands for, what the full APIPA address range is, and explain the exact condition under which Windows assigns an APIPA address instead of waiting for a DHCP server.
2. Rename the client machine following the same naming conventions researched in Section 2. After the rename and restart, verify and confirm the change using **three different methods** — document each method separately with a screenshot and its command or navigation path.
3. Assign the static IP `192.168.10.2`, subnet mask `255.255.255.0`, and default gateway `192.168.10.1` to the client. After applying, run `ipconfig /all` and produce a fully annotated screenshot in your report — label and explain every single field visible in the output without exception.
4. Open the Network Adapter Properties on the client and navigate to the **Advanced** tab of the IPv4 settings. Document every option visible. Specifically explain what the **Automatic Metric** setting does, what happens when you manually set it to a value of 1, and in what scenario an administrator would manually set this value.
5. Without changing any IP settings, **completely disable** the network adapter on the client from Device Manager. Run `ipconfig /all` and document the result. Re-enable the adapter and confirm the static IP configuration has been retained. In your report, explain the technical difference between disabling a network adapter and physically unplugging a network cable — what does each action affect at the OS and hardware level.
6. Open **Network and Sharing Center** on the client. Document the full network diagram shown and describe what each element in the diagram represents. Then click through to Change Adapter Settings and document all adapters listed — note their names, status, and type.
7. On the client, navigate to **Control Panel > System > Advanced System Settings > Environment Variables**. Document all **System Variables** currently listed. Identify and explain at least two variables that are directly relevant to networking or system identity, and explain what would happen if those variables were deleted or altered.

SECTION 4 — Advanced Connectivity Testing & Network Analysis

Marks: 30 | Estimated Time: 45 minutes

1. Before testing any connectivity, open **Windows Defender Firewall with Advanced Security** on the server. Navigate to Inbound Rules and locate the rules related to ICMPv4. Rather than disabling the entire firewall, enable only the specific ICMPv4 inbound rule required for ping. Document the exact rule name, the profile it applies to, and the protocol it controls. Perform the same action on the client. Explain why enabling a targeted rule is preferable to disabling the firewall entirely.
2. From the server, run a standard ping to the client's IP address. Document the full output and answer the following in your report: What does TTL mean? What is the default TTL value shown in the response? What does a lower TTL value indicate about the path a packet has taken in a multi-router network?
3. From the client, run the following four ping variations targeting the server. Document the complete output of each and write a clear explanation of what each variation tests and how it differs from the others:
   - j. **`ping 192.168.10.1`**
   - k. **`ping 192.168.10.1 -n 20`**
   - l. **`ping 192.168.10.1 -l 1000`**
   - m. **`ping 192.168.10.1 -w 1`**
4. From the server, run **`ping 192.168.10.2 -f -l 1472`** and document the result. Then increase the packet size to **`2000`** bytes and document the error message shown. In your report, explain what packet fragmentation is, what the **`-f`** flag prevents, and why large packets may need to be fragmented across a network.
5. Attempt to ping the client by its computer name from the server — document whether it succeeds or fails and explain why. Then open the **hosts file** located at **`C:\Windows\System32\drivers\etc\hosts`** on the server. Manually add an entry mapping the client's IP to its hostname, save the file, and attempt the name-based ping again. Document the result. In your report, explain what the hosts file is, the order of name resolution Windows follows, and how the hosts file relates to DNS.
6. After adding the hosts file entry, run both **`ping [ClientHostname]`** and **`nslookup [ClientHostname]`** from the server. Compare both results side by side in your report and explain the technical difference between how the hosts file resolves a name versus how DNS resolves a name.
7. On the client, run **`tracert 192.168.10.1`** and document every hop shown. In your report, explain how tracert differs from ping, what each column in the tracert output represents (hop number, three time columns, and hostname/IP), and what information a network administrator extracts from tracert output that is not available from a simple ping.
8. On the server, run **`arp -a`** before pinging the client — document the output. Then ping the client and immediately run **`arp -a`** again — document the changed output. Compare both results and explain: what the ARP table is, what it stores, how it was populated after the ping, and why ARP table entries expire over time.
9. On the server, run **`netstat -an`** and document all listening ports. Identify at least five ports, research what service or protocol each one corresponds to, and present your findings in a table. Run the same command on the client and identify at least two ports that appear on the server but not the client — explain why the server has these additional listening ports.

10. Deliberately introduce each of the following faults on the client **one at a time**. After each change, attempt a ping to the server and document the exact result and error message. Restore the correct setting completely before moving to the next fault. For each fault, write a technical explanation of why communication succeeded or failed:
    n. Change client IP to 192.168.20.2 (wrong subnet)
    o. Change subnet mask to 255.255.255.252
    p. Remove the Default Gateway entry entirely
    q. Set Default Gateway to 192.168.10.99 (unreachable gateway)
    r. Assign the same IP as the server: 192.168.10.1 (IP conflict)

> **NOTE:** Each fault in Task 10 must be fully restored before testing the next one. Running multiple faults simultaneously will produce ambiguous results and will not receive marks.

SECTION 5 — System Analysis, Event Investigation & Fault Diagnosis

Marks: 15 | Estimated Time: 35 minutes

Part A — Full System Profiling

1. On both the server and client, run **`systeminfo`** and copy the complete output of both into your report. Create a structured side-by-side comparison table covering: OS Name, OS Version, OS Build Type, Total Physical Memory, Available Physical Memory, Number of NICs, and total Hotfix count. Identify and explain at least four meaningful differences.
2. On both machines, open **`msinfo32`** and navigate to Components > Network > Adapter. Document the full adapter details for both machines in a comparison table. Then navigate to Software Environment > Network Connections and document all connections listed for each machine.
3. Use Task Manager on both machines simultaneously and record CPU, RAM, and network usage at the same moment in time for both. Document the results in a comparison table. Identify the top three memory-consuming processes on each machine — present their names, PIDs, memory usage, and CPU usage.

Part B — Service & Event Log Investigation

4. On the server, open Services (**`services.msc`**) and locate the **Windows Time** service. Document its current status, startup type, and logon account. Manually restart the service and document the process. Identify all services with Automatic startup type that are currently stopped — list them in a table and explain what this condition might indicate in a production environment.
5. Open Event Viewer on the server and navigate to Windows Logs > System. Filter to show only events from the last 24 hours. Document the total count of Information, Warning, and Error events. Identify one Warning or Error event — document its full details including Event ID, Source, Time, and Description, and research what that specific Event ID means.
6. Open Event Viewer > Windows Logs > Security on the server. Document the last five security audit events. Identify each Event ID, explain what each one represents in terms of security activity, and explain why security teams actively monitor the Security log in enterprise environments.
7. Deliberately attempt to log in to the server using a wrong password **three times in succession**. Then open Event Viewer > Security log and locate the failed login events generated. Document the Event ID, the full event description, and the username and machine name recorded in the event. Explain what information this log entry contains and why organizations use this to detect unauthorized access attempts.

Part C — Final Fault Diagnosis Scenario

> **INSTRUCTION:** This scenario tests your ability to independently diagnose and repair a broken server environment using only the command line. You will introduce all faults yourself, then fix them systematically using only CLI tools — no GUI allowed during the repair phase.

8. Introduce ALL of the following changes simultaneously on the client machine, then repair each problem using only command-line tools. Document every diagnostic command you run, its full output, and your reasoning at each step as if writing a professional IT incident report:
   s. IP address changed to 192.168.50.10
   t. Subnet mask changed to 255.255.0.0

u. Default gateway removed completely
v. Computer renamed using characters that violate naming rules
w. Time zone changed to a completely incorrect zone (e.g., UTC+12)
x. DNS server on the client changed to 8.8.8.8

After repairing all issues using only the command line, verify full connectivity is restored by running a final ping from the client to the server and from the server to the client. Document every step of your diagnosis and resolution as a formal incident report.

LAB REPORT SUBMISSION REQUIREMENTS

Marks: Embedded in section scores — report quality directly affects your total

Your complete lab report must be submitted as a single structured document immediately after the assessment concludes. The report must follow the structure defined below. Reports that do not follow this structure will be penalized.

1. Cover Page

Your report must open with a cover page containing: your full name, registration number, section/batch, date of assessment, the hostnames you assigned to both machines, and the IP addresses you configured on both machines.

2. Environment Summary Table

Immediately after the cover page, include a completed configuration summary table. This table must reflect the final verified state of both machines at the end of the assessment.

Configuration Item	Server Value	Client Value
Hostname		
IP Address		
Subnet Mask		
Default Gateway	N/A	
DNS Server		
Time Zone		
MAC Address		
OS Version		
Snapshot Name (Pre-config)		

3. Section-by-Section Task Responses

For each section of this assessment, your report must contain a dedicated chapter. Within each chapter, address every task in order. For each task include: a numbered heading matching the task number, one or more annotated screenshots as required, and your written explanation or analysis in complete sentences.

Report Section	Content Required	Minimum Length
Section 1	VM comparison table, NAT vs Internal test analysis, host resource analysis, MAC address documentation, snapshot procedure	1 page minimum

Report Section	Content Required	Minimum Length
Section 2	Server Manager field table, naming convention research, command outputs with annotations, time sync analysis, advanced IP settings exploration	2 pages minimum
Section 3	APIPA explanation, rename verification, fully annotated ipconfig output, adapter properties analysis, environment variables documentation	1.5 pages minimum
Section 4	Firewall rule documentation, TTL analysis, ping variation results, fragmentation test, hosts file experiment, tracert analysis, ARP table comparison, netstat table, fault diagnosis table	2.5 pages minimum
Section 5	Systeminfo comparison table, msinfo32 documentation, service analysis, event log findings, fault diagnosis incident report	2 pages minimum

4. Screenshot Standards

Every screenshot submitted must meet the following standards: the full VM window must be visible (not just a cropped portion), the taskbar clock must be visible to confirm the time of capture, screenshots must be in the correct sequence matching the task order, and each screenshot must be followed immediately by a caption identifying the task number and what is being shown.

5. Written Explanation Standards

All explanations must be written in your own words. Answers that are identical or closely paraphrased from online sources will receive zero marks for that task. Explanations must be technically accurate, specific to what you observed on your own machines, and relevant to the task question asked.

6. Challenges & Errors Log

At the end of each section in your report, include a short paragraph titled Challenges & Errors. In this paragraph, document every unexpected result, error message, or difficulty you encountered during that section and describe how you resolved it. Students who document no challenges at all will be questioned about their report in a verbal follow-up.

7. Final Reflection

At the very end of the report, include a reflection of no less than two paragraphs. In the first paragraph, describe the three most important technical concepts you reinforced or learned during this assessment. In the second paragraph, describe one mistake you made, what it caused, and how diagnosing and fixing it helped your understanding of Windows Server administration.

ACADEMIC INTEGRITY DECLARATION

By submitting this report, I confirm that all work, screenshots, command outputs, and written explanations contained herein are entirely my own. I have not copied any content from external sources, other students, or AI-generated text. I understand that any evidence of academic dishonesty will result in a score of zero for the entire assessment.

Student Signature: ______________________________ **Date:** _____________